

YUSHI WANG

EXPERIENCE DESIGNER · DESIGN TECHNOLOGIST — AEC · BIM · AI · FULL-STACK

yushiw620@gmail.com · Boston, MA (+1) 213-706-9087 · Shanghai (+86) 139-1616-4162

PROFILE

Background	Sole design technologist at a 200-person global AEC firm. Partners directly with the Principals who lead the Building Science, Space Strategy and Healthcare practices, turning the needs of each discipline into the right tool, and works the day-to-day details with the designers and Senior Associates who use it. Owns the full path: research, interaction design, testing and detail work, design specs, release, then usage tracking and internal rollout. Effectively a one-person product company serving the whole firm.
Experience Design	Runs the full design arc for technical AEC and BIM tools. Turns user research, including interviews, workflow analysis and qualitative and quantitative methods, into concrete product improvements. Designs and weighs several options, prototypes from low to high fidelity, and validates each with real users and the engineers who build it. Usability testing and behavioral data decide what ships, with the chosen design carried through to working software, build issues included.
Impact	More than a decade in the Autodesk AEC stack across Revit, AutoCAD and Dynamo, with a working knowledge of the BIM workflows the industry actually runs on. That fluency turned into products: a sustainability platform that won the AIA TAP Innovation Award and reached users in 85+ countries, and an AI rendering pipeline that sped up image output ~70% and cut \$50,000+ of outsourcing.
Enablement	Founded and led an internal AI research group and a 100+ member Computational Design Community. The work grew from one-off scripts and tools into structured teaching, then into AI tools that capture institutional knowledge and put it back to work across AIGC, parametric design and marketing. Every request is treated as a product question: the smallest set of features that helps the most people, wired into a loop that keeps improving and steadily lifts firm-wide digital maturity.
Profile	Goal-oriented and detail-minded. Cuts quickly to the core of a messy problem, names the real bottleneck and user pain, and pushes the fix to a shipped release. Leverages vibe coding to turn new requirements and ideas into validated prototypes within days, sharply compressing the concept-to-build cycle. Communicates clearly in writing, in person and in visuals, works in fluent English, and picks up new tools and domains fast.

SELECTED PROJECTS

Interactive Virtual Mockup (IVM) <i>Experience Design · Immersive VR / AR</i> Role Physical healthcare mockups cost \$12,000–\$18,000 per room and cannot show movement, workflow or fast iteration. Built a multi-user VR tool where surgeons, nurses, infection-control staff and architects walk a clinical room before construction, operate real equipment, switch design options, and give structured feedback while their real workflows are observed in context. Build Pipeline from Rhino and Revit into Unity and onto Meta Quest, shipped over three releases: an interactive base environment with screenshot and voice annotation, then higher-fidelity lighting with button-driven equipment controls, then synchronized multiplayer review of complex multi-layer ceilings. An AR track places full-scale options on the real site and swaps components on physical models. Impact Saved \$112,000 across seven mockup rooms at Dana-Farber Cancer Institute and cut medical-gas change orders 60% by catching clearance and workflow conflicts before ground-breaking, compressing the cross-disciplinary feedback cycle from weeks to days. Adopted on White Plains Hospital and Dana-Farber; the Chief of Surgery at White Plains called it the most significant advancement in surgical facility design they had seen, and it drew a strong response at a national healthcare design conference.	2024 — 2025
Floorcast <i>Experience Design · AI Building-Use Simulation</i> Role Predicts how a building will really be used before it is built: AI agents play each clinical role to reproduce real routes and workflows and expose how a layout performs. Build Movement, comfort and acoustic heatmaps with live data give the quantitative picture; an AI interview system adds the qualitative why. Extending from one-click 2D-to-3D to a collaborative Revit and Rhino plugin that runs simulation inside the design workflow, toward an AI-agent, data-driven platform that predicts how a building will perform before it is built. Impact Analyzed nurse call-response and rounding times and routes to test nurse-station placement, then re-strategized the wall and double-door positions of the equipment and medication rooms to shorten clinical circulation paths.	Feb 2026 — Present
PAYETTE Places <i>Experience Design · Interactive Web</i> Role Designed the navigation and wayfinding experience for the firm public project tour at the 2025 AIA Conference, a design-communication artifact serving a broad external audience of architects, clients and press across eight built Boston projects. Build Information architecture spanning an interactive city map with one-tap Google Maps routing, building hotspots that open detailed project pages, a navigable gallery with info panels, and animated facade diagrams with walking figures and interactive envelope details; fully responsive, with the site QR printed on company business cards. Impact Became the public-facing showcase for AIA 2025 and a benchmark for how the office presents its work, with roughly 300 visitors on a single open day.	Mar — Jun 2025
Promptitex <i>Product & Experience Design · AI Prompt Intelligence</i> Role An AI prompt-intelligence layer embedded in the design process that turns scattered, one-off AI experiments into a shared, continuously improving capability. Build A searchable knowledge gallery, RAG case retrieval with match scores, a guided prompt builder, and a node-based board for comparing model outputs. Shaped by how designers actually work, it is tuned so someone with no AI background gets reliable results in fewer steps, with design copyright and IP compliance built into the workflow. Impact Standardized prompt quality across AIGC, parametric design and marketing, raised output consistency, and protected design copyright while lowering the barrier for everyday designers.	Sep 2025 — Present

PROFESSIONAL EXPERIENCE

PAYETTE — Boston, MA

June 2022 — Present

Design Technologist

- **Validation & Feedback** Built the client-feedback loop that optimizes the design process before construction: Pulse, a web tool that collects structured feedback on 2D plans and digital spatial-design options, and immersive AR/VR review in Unity, C# and iOS ARKit that lets clinical, engineering and architecture teams validate space and equipment decisions on site.
- **User Research & Experience** Own user research and interaction design for the firm internal tools serving 200+ architects, engineers and BIM staff: space-strategy tools (Floorish, Sectioneer, Floorcast), sustainability tools (Kaleidoscope, an embodied-carbon platform that won the AIA TAP Award and reached 85+ countries; Solar Comfort) and the IVM healthcare VR review tool (\$112K mockup savings, change orders down 60%). Conduct interviews, workflow analysis and contextual observation, build low- to high-fidelity prototypes, and lead usability testing to validate current and future features.
- **Computational Design** Build professional AEC tooling hands-on: 30+ Grasshopper and Dynamo scripts, a .NET Revit add-in that installs a company plugin ribbon of in-house tools across the office, plus Python Rhino plugins and generative design; consolidated into the firm parametric script library and digital training system, with regular digital and AI lectures that brought 100+ designers to frontier practice.
- **Applied AI** Drove applied AI end to end: led the AI research group that tuned image-to-image generation into a faster rendering pipeline (~70% efficiency, \$50K+ outsourcing saved), built a prompt-intelligence tool that makes reliable image generation easy for any designer and an AI video tool that turned competitive project-bid pitches from static decks into dynamic visual storytelling, and now use vibe coding to stand up MVP demos in days plus AI agents, LLMs and RAG to ship internal web tools that streamline the design workflow.
- **Full-Stack Delivery** Take 10+ tools from concept to production largely single-handed: meet regularly with the three practice Principals and each project-group PM to surface needs and pain points and turn them into solutions, build front and back end, deploy on Azure with versioned releases, and partner with engineering to land design specs and resolve implementation issues, lifting target workflows 50 to 80 percent. Partnering with the marketing and finance teams, also built the Ask Tom RAG agent and Payette Lens, an AI financial-data visualization tool.
- **Leadership & Management** Founded and lead the firm AI research group and a 100+ member design-tech community: set multi-year tooling and AI strategy, build training, onboarding and version-controlled knowledge systems, govern design and IP standards, mentor designers across offices, report monthly to firm Principals and leadership, and drive cross-functional, globally distributed collaboration that influences decisions at every level.

Dowbuilt — Seattle, WA

Dec 2021 — June 2022

Digital Construction Project Engineer

- **BIM Automation** Built a Python/Revit BIM modeling and automated quantity-take-off system; one-click material lists and cost estimates cut take-off from days to minutes (about 90% faster), with real-time cost data across 10+ high-end residential projects.
- **Pre-construction & Field** Delivered Grasshopper/Dynamo/Python tooling for excavation modeling, 4D scheduling, clash detection and CNC-ready fabrication; coordinated weekly with PMs, superintendents and fabricators.

INTERNSHIP

NBBJ — Seattle, WA

July — Sept 2021

Digital Design Specialist Intern

- **UI/UX** Led a full UI/UX redesign of the firm enterprise digital-technology platform (unified tool search and technical-asset management), improving usability and cross-office collaboration.
- **Parametric R&D** Built parametric Grasshopper/Dynamo tools (brick-porosity, solar-tracking façade, automated daylight simulation) adopted in pilot projects and QA/QC.

EDUCATION

University of Southern California

M.S. Building Science · GPA 4.0 / 4.0

Los Angeles · 2019 — 2021

Shanghai University of Engineering Science

B.S. Construction Management · GPA 4.0 / 4.0

Shanghai · 2016 — 2018

SKILLS & CERTIFICATE

DESIGN Figma · Adobe Creative Cloud · InDesign · Claude Design

COMPUTATIONAL DESIGN Rhino · Grasshopper · Revit · Dynamo · Rhino.Inside

DEVELOPMENT JavaScript · HTML/CSS · Node.js · React · Python · GitHub · Azure · Unity · C# · iOS ARKit · Normcore · MySQL · Power BI · Tableau · D3 · Power Automate

AI Claude Code · Codex · Gemini · ComfyUI · xfigura · Krea · Runway · Kohya · RAG

CERTIFICATE LEED AP (USGBC, 2021)

AWARDS 2022 ARCHITECT R+D Award & AIA TAP Innovation Award (Kaleidoscope) · IESLA Russell Cole Memorial Design Competition — Special Recognition, Advanced (2020)